

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| Semester             | B.E. Semester VII– Electronics and Computer Science |
| Subject              | DAV Laboratory                                      |
| Laboratory Professor | Prof. Uma Jaishankar                                |

|              |                                                                         |
|--------------|-------------------------------------------------------------------------|
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| Roll Number  | 23108B0084                                                              |
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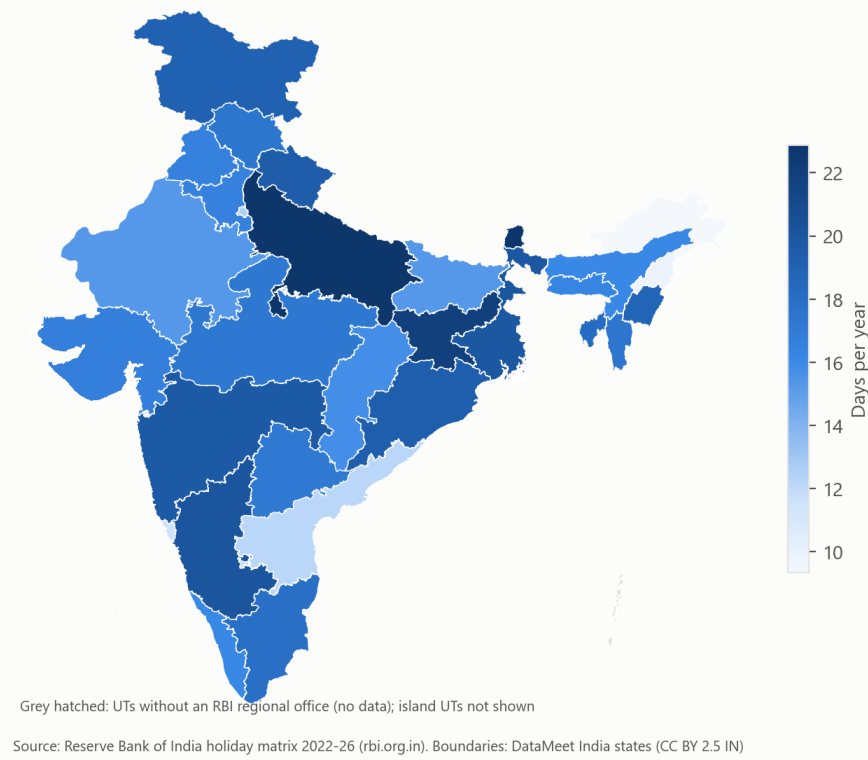
|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mini Project      | <b>Socio-Economic &amp; Demographic Analysis of Indian Festivals</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Problem Statement | India exhibits immense cultural and regional diversity, with hundreds of festivals celebrated across states, communities and calendar months. Unstructured information leads to biased representations and missing comparative insight. The project must (1) collect an authentic multi-faith festival dataset with zero synthetic records and exact source URLs, (2) build and evaluate ML models, (3) perform rigorous statistical testing and EDA, (4) publish a public data-visualisation blog with 40+ external interactions, and (5) produce a synthesis video.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Resources         | Software: Python 3.12, Windows 11, ffmpeg, Windows SAPI (System.Speech) for narration, Microsoft Word (PDF export), MS Edge + Playwright.<br>Libraries: pandas, numpy, scipy, statsmodels, scikit-learn, joblib, matplotlib, plotly, requests, BeautifulSoup, xldr, markdown.<br>Data sources (all real, cited per record): Reserve Bank of India holiday matrix 2022-26 (rbi.org.in); DoPT compulsory-holiday list (O.M. F.No.12/2/2023-JCA); Census of India 2011 Table C-01 (censusindia.gov.in); Wikipedia infoboxes via MediaWiki API + Govt. portals (utsav.gov.in, soreng.nic.in, dtahills.mn.gov.in, megtourism.gov.in); CAIT (cait.in) and verified news for spending; NRF, gov.cn, GASTAT, muenchen.de for global analogues; DataMeet state boundaries (CC BY 2.5 IN).                                                                                                                                                                                                                                                                                                                |
| Deliverables      | Task 1 - data/raw/*.csv json (raw with source columns), data/processed/*.csv (clean), data/data_dictionary.md csv<br>Task 2 - code/06_ml_models.py, model weights outputs/models/*.joblib, metrics outputs/ml/*.csv, ml_summary.json<br>Task 3 - outputs/stats/statistical_summary.md + T1-T10 validation tables, figures F01-F10<br>Task 4 - blog/blog_post.md html, blog/images, blog/interactive/*.html, blog/engagement (log template, audit report)<br>Task 5 - video/Indian_Festivals_Project_Report.mp4 (1080p, narrated), video/narration_script.md, this report                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Approach          | 1. Scrape the RBI holiday matrix for every office and month of 2022-2026 (3,221 office-date rows), save raw HTML + CSV with source URL per row.<br>2. Build a 101-festival regex catalog and resolve each RBI label into festivals. Shared dates are credited by evidence: a festival counts in full where the office is confirmed to observe it (closed on a date where it stood alone, on the DoPT compulsory list and never absent, or the festival's home community); the rest is split by an evidence score and flagged. Sunday correction: each festival is averaged only over the years it could be listed.<br>3. Source-ground each festival's tradition from its Wikipedia infobox ('Observed by') or an official Govt. page.<br>4. Download Census 2011 C-01, map 34 RBI offices to 29 states/UTs, build festival-, observation- and state-level tables.<br>5. Collect 78 economic/football figures; open every page to confirm the figure (snippet-only figures were excluded).<br>6. EDA & hypothesis tests (chi-square, permutation, Spearman, Kruskal-Wallis, Mann-Whitney, OLS). |

|                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |                        |                        |               |          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------|------------------------|---------------|----------|
|                                                                                                                                                                                                                                                                                                                                           | 7. ML: K-Means/Ward clustering of states, Random Forest vs Logistic tradition classifier, log-linear trade regression with LOO and temporal hold-out.<br>8. Blog + interactive Plotly charts, engagement audit tooling; 9. narrated video; 10. report.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |                        |                        |               |          |
| Code                                                                                                                                                                                                                                                                                                                                      | <div>01_scrape_rbi_holidays.py02_verify_wikipedia.py03_economic_footfall_records.py</div> <div>04_build_clean_dataset.py05_eda_statistics.py06_ml_models.py</div> <div>07_data_dictionary.py08_interactive_charts.py09_blog_and_engagement.py</div> <div>10_make_video.py11_build_report.pyfestival_catalog.pyviz_style.py</div> <div>Excerpt - fractional attribution of shared-date holidays (04_build_clean_dataset.py):</div> <div>obs["n_items_on_date"] = obs.groupby(["date", "rbi_regional_office"])["festival"]</div> <div>.transform("size") + obs["n_civic_items"]</div> <div>obs["attribution_weight"] = 1 / obs["n_items_on_date"]</div> <div>obs["attribution"] = np.where(obs["n_items_on_date"] == 1, "unique", "shared_date")</div> |           |                        |                        |               |          |
| Task 1: Dataset                                                                                                                                                                                                                                                                                                                           | 3,271 festival-holiday observations   101 festivals   34 RBI offices   29 states/UTs   2022-2026   78 verified economic/footfall records   Census 2011 demography. Zero synthetic records.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |           |                        |                        |               |          |
|                                                                                                                                                                                                                                                                                                                                           | tradition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | festivals | holiday-days/office/yr | % of festival holidays | % moving-date |          |
|                                                                                                                                                                                                                                                                                                                                           | Hindu                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 45        | 8.58                   | 49.4                   | 77.78         |          |
|                                                                                                                                                                                                                                                                                                                                           | Tribal/Indigenous                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 23        | 0.91                   | 5.2                    | 60.87         |          |
|                                                                                                                                                                                                                                                                                                                                           | Buddhist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 7         | 0.69                   | 4.0                    | 71.43         |          |
|                                                                                                                                                                                                                                                                                                                                           | Muslim                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 7         | 3.41                   | 19.6                   | 100.0         |          |
|                                                                                                                                                                                                                                                                                                                                           | Christian                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 7         | 2.07                   | 11.9                   | 14.29         |          |
|                                                                                                                                                                                                                                                                                                                                           | Sikh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 5         | 0.73                   | 4.2                    | 80.0          |          |
|                                                                                                                                                                                                                                                                                                                                           | Cultural (multi-faith)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4         | 0.36                   | 2.1                    | 25.0          |          |
|                                                                                                                                                                                                                                                                                                                                           | Jain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2         | 0.53                   | 3.1                    | 100.0         |          |
| Parsi                                                                                                                                                                                                                                                                                                                                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.09      | 0.5                    | 0.0                    |               |          |
| Limitations: 19 of 101 festivals always share their date, so their state coverage stays uncertain (flagged); festivals without a bank holiday, restricted and district-level holidays and UTs without an RBI office are not covered; bank holidays measure official recognition, not participation; trade figures are industry estimates. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |                        |                        |               |          |
| Task 3: Statistics                                                                                                                                                                                                                                                                                                                        | variable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | n         | mean                   | median                 | std           | skewness |
|                                                                                                                                                                                                                                                                                                                                           | festival_expected_states_per_year                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 101.0     | 4.29                   | 1.03                   | 7.0           | 2.22     |
|                                                                                                                                                                                                                                                                                                                                           | festival_mean_holiday_days (duration)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 101.0     | 1.05                   | 1.0                    | 0.18          | 4.11     |
|                                                                                                                                                                                                                                                                                                                                           | festival_date_drift_days (moving festivals)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 69.0      | 31.65                  | 22.0                   | 58.01         | 5.38     |
|                                                                                                                                                                                                                                                                                                                                           | state_festival_holiday_days_per_year (frequency)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 29.0      | 17.01                  | 17.23                  | 3.44          | -0.5     |
|                                                                                                                                                                                                                                                                                                                                           | state_holiday_diversity_shannon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 29.0      | 1.33                   | 1.3                    | 0.18          | 0.46     |
|                                                                                                                                                                                                                                                                                                                                           | state_distinct_festivals_2024_26                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 29.0      | 14.59                  | 15.0                   | 2.99          | -0.78    |
|                                                                                                                                                                                                                                                                                                                                           | festive_trade_INR_crore (spending)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 18.0      | 118166.67              | 37500.0                | 173144.7      | 1.77     |
|                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |                        |                        |               |          |
|                                                                                                                                                                                                                                                                                                                                           | id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | test      | statistic              | p_value                | result        |          |
| H1                                                                                                                                                                                                                                                                                                                                        | Chi-square goodness-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 825.499   | < 0.001                | Reject H0              |               |          |

|                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |                      |                   |                   |
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|                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | of-fit (12 months, weighted                      |                      |                   |                   |
|                                                                                                                                                                                                                                                                                                                                                                                                            | H2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Chi-square test of independence (+5,000-permutat | 5.691                | 0.785             | Fail to reject H0 |
|                                                                                                                                                                                                                                                                                                                                                                                                            | H3-Hindu                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Spearman rank correlation (n=29 states/UTs)      | 0.614                | < 0.001           | Reject H0         |
|                                                                                                                                                                                                                                                                                                                                                                                                            | H3-Muslim                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Spearman rank correlation (n=29 states/UTs)      | 0.384                | 0.040             | Reject H0         |
|                                                                                                                                                                                                                                                                                                                                                                                                            | H3-Christian                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Spearman rank correlation (n=29 states/UTs)      | 0.495                | 0.006             | Reject H0         |
|                                                                                                                                                                                                                                                                                                                                                                                                            | H3-Sikh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Spearman rank correlation (n=29 states/UTs)      | 0.642                | < 0.001           | Reject H0         |
|                                                                                                                                                                                                                                                                                                                                                                                                            | H3-Buddhist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Spearman rank correlation (n=29 states/UTs)      | 0.645                | < 0.001           | Reject H0         |
|                                                                                                                                                                                                                                                                                                                                                                                                            | H3-Jain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Spearman rank correlation (n=29 states/UTs)      | 0.662                | < 0.001           | Reject H0         |
|                                                                                                                                                                                                                                                                                                                                                                                                            | H4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Kruskal-Wallis H (identifiable-reach festivals)  | 10.283               | 0.016             | Reject H0         |
|                                                                                                                                                                                                                                                                                                                                                                                                            | H5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mann-Whitney U                                   | 870.500              | 0.708             | Fail to reject H0 |
|                                                                                                                                                                                                                                                                                                                                                                                                            | H6-hol_days                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Kruskal-Wallis H                                 | 2.362                | 0.797             | Fail to reject H0 |
|                                                                                                                                                                                                                                                                                                                                                                                                            | H6-holiday_                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Kruskal-Wallis H                                 | 6.225                | 0.285             | Fail to reject H0 |
|                                                                                                                                                                                                                                                                                                                                                                                                            | H7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Spearman rank correlation                        | 0.314                | 0.097             | Fail to reject H0 |
|                                                                                                                                                                                                                                                                                                                                                                                                            | H8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mann-Whitney U on log trade (national CAIT-type  | 52.000               | 0.109             | Fail to reject H0 |
|                                                                                                                                                                                                                                                                                                                                                                                                            | H9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | OLS log(trade) ~ year + festival fixed effects   | 12.469               | < 0.001           | Reject H0         |
| <p>H1: holidays cluster by month (chi2 = 825, p &lt; 0.001; peaks Mar, Oct, Apr). H2: season independent of tradition (perm. p = 0.78). H3: holiday share correlates with population share for all six faiths (rho 0.38-0.66, all p &lt; 0.05 after Holm correction). H4: reach differs by tradition group (p = 0.016). H5-H8 not significant. H9: festive trade grows 34.0%/yr (95% CI 27.2 to 41.1).</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |                      |                   |                   |
| Task 2: Machine Learning                                                                                                                                                                                                                                                                                                                                                                                   | <p><b>A. Clustering states (K-Means vs Ward)</b><br/>Features: Hindu/Muslim/Christian/Tribal shares of festival holidays (chosen by a documented feature-set experiment). k = 2 (silhouette 0.502); bootstrap ARI 0.68 +/- 0.33; ARI vs region 0.01.<br/>C1: typical mix: Andhra Pradesh, Arunachal Pradesh, Assam, Bihar, Goa, Gujarat, Himachal Pradesh, Jharkhand, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Odisha, Punjab-Haryana-Chandigarh, Rajasthan, Sikkim, Tamil Nadu, Telangana, Tripura, Uttar Pradesh, Uttarakhand, West Bengal; C2: more Tribal/Indigenous &amp; Christian holidays than average: Manipur, Meghalaya, Mizoram, Nagaland; C3: more Muslim holidays than average: Chhattisgarh, Delhi, Jammu &amp; Kashmir (the 3-group view, silhouette 0.41).</p> |                                                  |                      |                   |                   |
|                                                                                                                                                                                                                                                                                                                                                                                                            | <p><b>B. Tradition classifier (5-fold x 20 repeated stratified CV)</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                  |                      |                   |                   |
|                                                                                                                                                                                                                                                                                                                                                                                                            | model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | accuracy_mean                                    | macro_precision_mean | macro_recall_mean | macro_f1_mean     |
|                                                                                                                                                                                                                                                                                                                                                                                                            | Majority baseline                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0.446                                            | 0.111                | 0.25              | 0.154             |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |      |       |       |       |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|-------|-------|-------|
|                           | Logistic Regression                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0.48  | 0.45 | 0.455 | 0.429 | 0.097 |
|                           | Random Forest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0.514 | 0.47 | 0.468 | 0.446 | 0.1   |
|                           | <b>C. Festive-trade regression (log-linear, festival fixed effects)</b><br>n = 15; in-sample R <sup>2</sup> (log) 0.993; LOO RMSE 0.281 log units; LOO MAPE 20.2%; growth 35.5%/yr; hold-out 2026 MAPE 13.1% (Holi 2.9%, Raksha Bandhan 23.2%).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |      |       |       |       |
| Task 4: Blog & Engagement | Article 'What India's Bank-Holiday Calendar Reveals About How the Country Celebrates' (~1,700 words, 9 figures, 8 footnoted sources, secular framing) in blog/blog_post.md and a paste-ready blog_post.html; 4 interactive Plotly charts (hover tooltips) in blog/interactive/.<br>Publication status: <a href="https://medium.com/@i.jainamjain/what-indias-bank-holiday-calendar-reveals-about-how-the-country-celebrates-97c99917b2d3">https://medium.com/@i.jainamjain/what-indias-bank-holiday-calendar-reveals-about-how-the-country-celebrates-97c99917b2d3</a> . External interactions logged: 0 / 40 target; comments answered: 0. The audit (09_blog_and_engagement.py) counts only people outside VIT and requires a screenshot per row. Publishing and community engagement must be done by the author; no engagement is simulated. |       |      |       |       |       |
| Task 5: Video             | video/Indian_Festivals_Project_Report.mp4 - 1920x1080, H.264/AAC, 11 narrated slides (~5.4 min): motivation & source integrity, statistics, ML architecture & performance, economics, blog & reception, takeaways. The reception slide reads engagement_metrics.json; re-run 10_make_video.py after publishing to produce the final cut.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |      |       |       |       |
| Output Visuals            | <p>Source: Reserve Bank of India holiday matrix 2022-26 (rbi.org.in). Sunday-corrected; shared dates credited by evidence score.</p> <p><b>Fig 1. Festival bank-holiday days by month and tradition (RBI 2022-26).</b></p> <p><b>State-wise festival heatmap: when each state's holidays fall</b></p> <p>Fig 2. State x month heatmap.</p> <p>Source: Reserve Bank of India holiday matrix 2022-26 (rbi.org.in)</p>                                                                                                                                                                                                                                                                                                                                                                                                                             |       |      |       |       |       |

**Festival bank holidays per year, by state**



**Fig 3. Festival holidays per year by state.**

Do holiday calendars mirror demography? (each dot = one state; dashed = trend)

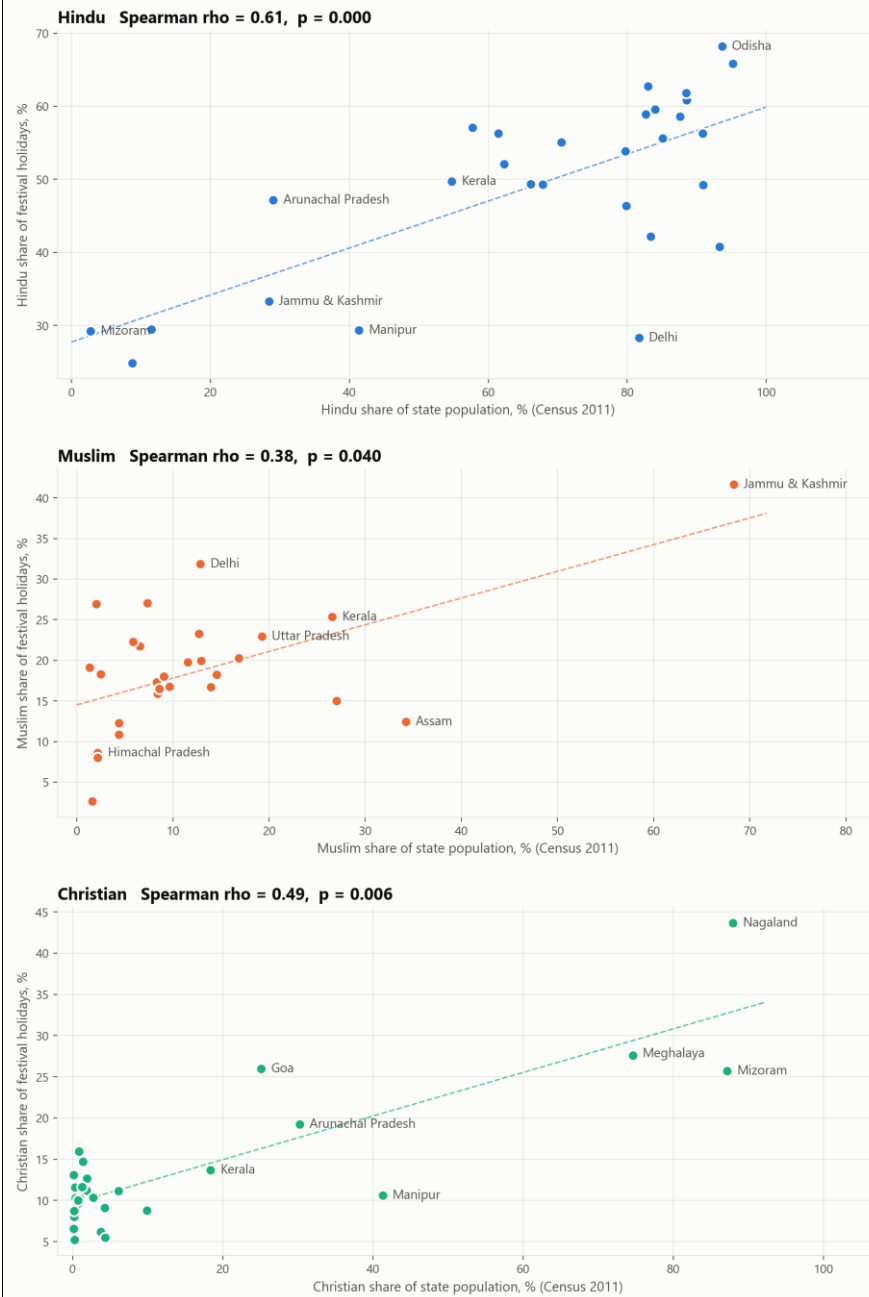


Fig 4. Holiday share vs Census population share.

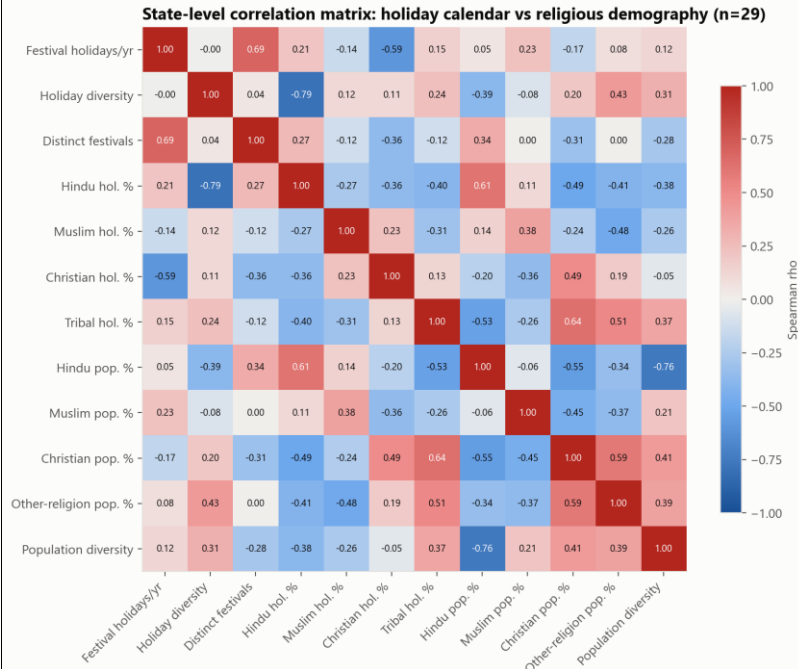


Fig 5. State-level Spearman correlation matrix.

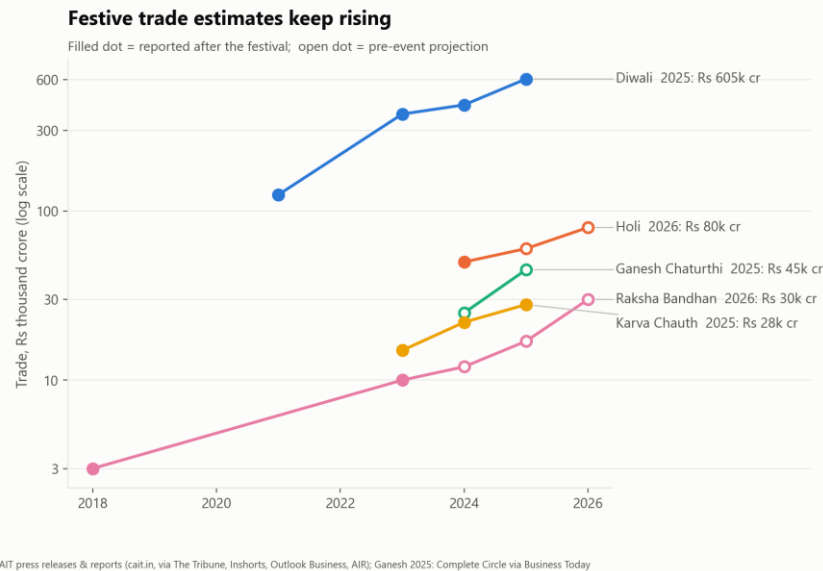


Fig 6. Festive trade estimates (CAIT).

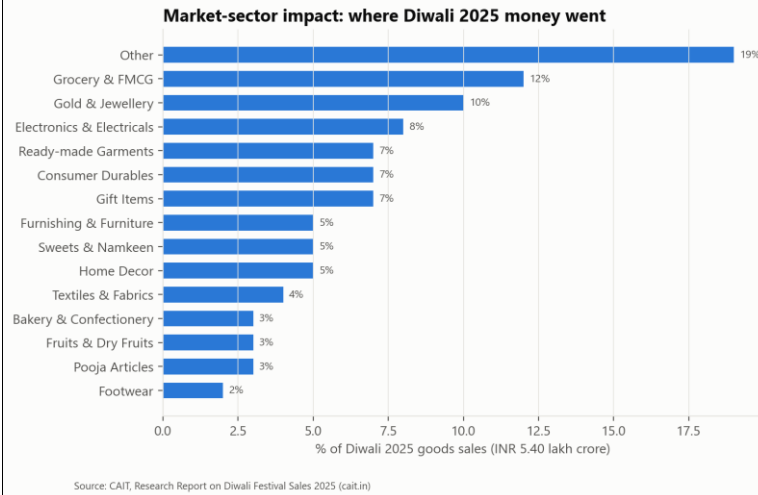


Fig 7. Diwali 2025 sector split.

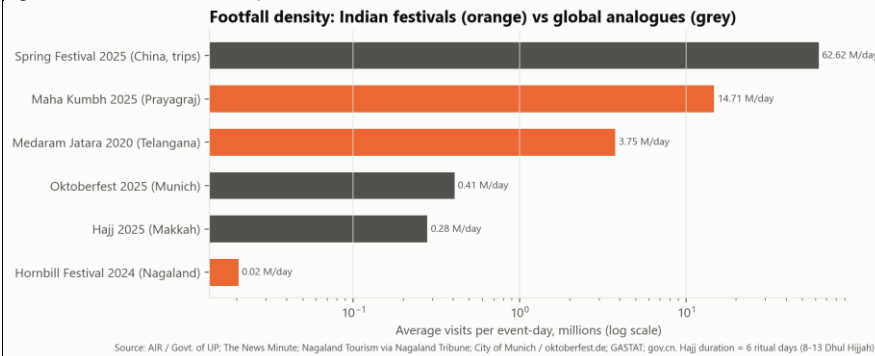


Fig 8. Footfall density vs global analogues.

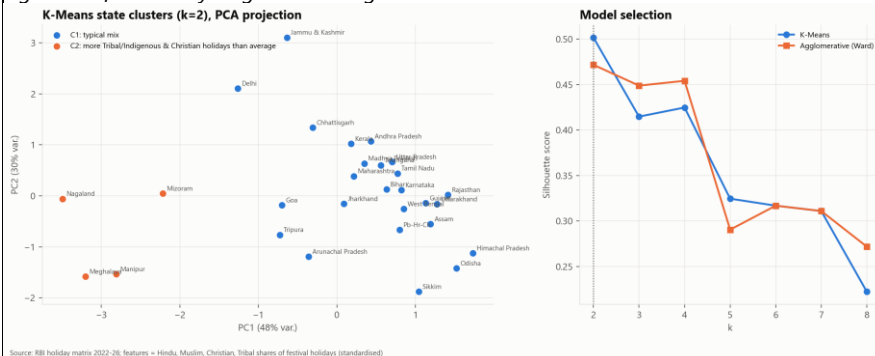
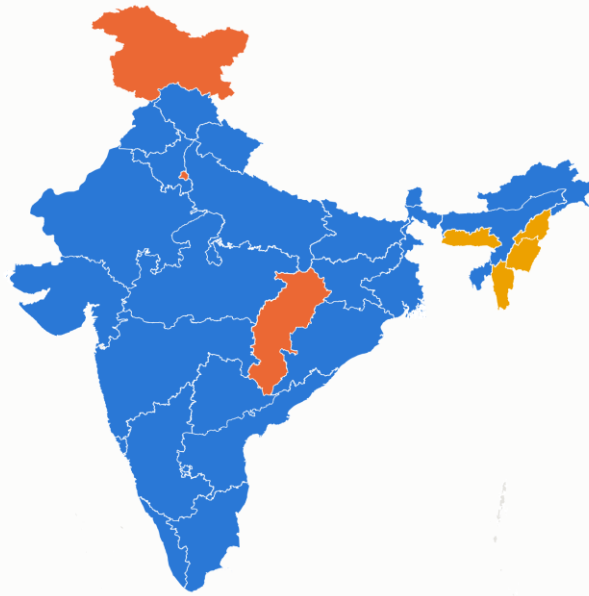


Fig 9. K-Means state clusters (PCA) and model selection.

### Three kinds of festival calendar in India

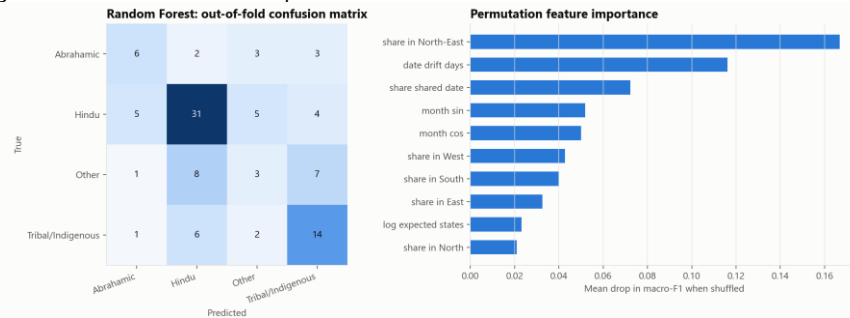
K-Means on each state's Hindu, Muslim, Christian and tribal holiday shares. Strongest split: 2 groups (silhouette 0.50); this 3-group view scores 0.41



- Typical mix (close to the all-India average) (22 states)
- More Tribal/Indigenous & Christian holidays than average (4 states)
- More Muslim holidays than average (3 states)
- No RBI office (no data)

Source: RBI holiday matrix 2022-26; boundaries DataMeet (CC BY 2.5 IN); island UTs not shown

Fig 9b. State clusters on the map.



Source: Festival features derived from RBI holiday matrix 2022-26; labels from Wikipedia / Govt. attribution (festival\_mastecss)

Fig 10. Classifier confusion matrix and permutation importance.

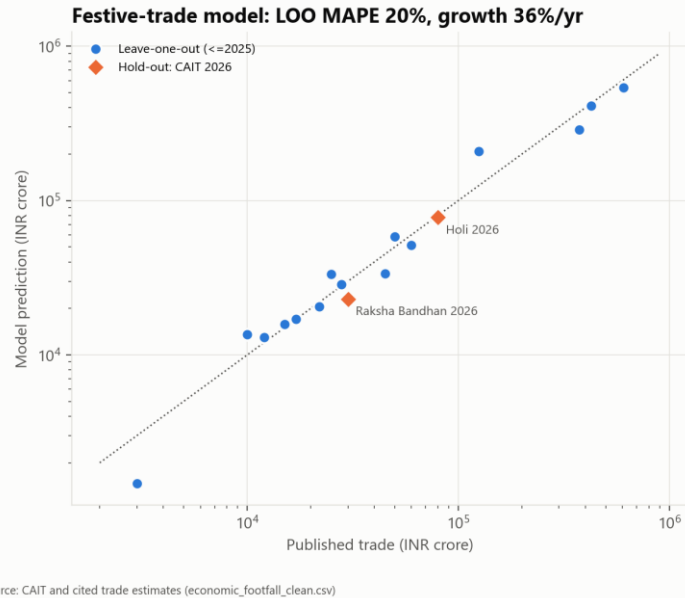


Fig 11. Trade model: LOO and 2026 hold-out predictions.

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Insights   | <ol style="list-style-type: none"> <li>1. Festival holidays peak in April, October and March (about 2.6-2.8 days per office each); January, February and July are quietest.</li> <li>2. Every faith's holiday share rises with its population share (Spearman 0.38-0.66), yet the calendar is more plural than the population: Hindu festivals are ~49% of festival holidays vs 79.8% of population, because Christmas, Good Friday, both Eids, Buddha Purnima, Mahavir Jayanti and Guru Nanak Jayanti are gazetted almost nationwide.</li> <li>3. Uttar Pradesh (22.9), Sikkim (22.8) and Jharkhand (22.0) have the most festival holidays; Arunachal Pradesh (9.3) the fewest. Regions do not differ significantly.</li> <li>4. The strongest state grouping is 2 clusters (four north-eastern hill states vs the rest, silhouette 0.50); calendars otherwise form a continuum that does not follow regions (ARI 0.01).</li> <li>5. A festival's date and footprint only weakly reveal its faith (macro-F1 0.45): nationwide festivals of all faiths look alike.</li> <li>6. Reported festive trade grows ~34%/yr (Diwali: Rs 1.25 lakh cr in 2021 -&gt; Rs 6.05 lakh cr in 2025), but national spending estimates exist mainly for Hindu festivals - a measurement blind spot.</li> </ol> |
| Conclusion | A fully source-grounded, multi-faith festival dataset was built from official RBI and Census data plus 78 verified economic figures, analysed with hypothesis tests and three ML models, and communicated through a footnoted blog, interactive charts and a narrated video. Remaining author actions: publish the article, collect 40+ genuine external interactions, fill the engagement log, and re-run the audit and video scripts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| References | <ol style="list-style-type: none"> <li>[1] RBI, Holidays under NI Act - <a href="https://www.rbi.org.in/Scripts/HolidayMatrixDisplay.aspx">https://www.rbi.org.in/Scripts/HolidayMatrixDisplay.aspx</a></li> <li>[2] Census of India 2011, C-01 - <a href="https://censusindia.gov.in/nada/index.php/catalog/11361">https://censusindia.gov.in/nada/index.php/catalog/11361</a></li> <li>[3] CAIT Diwali 2025 report - <a href="https://cait.in">https://cait.in</a> (10 Nov 2025); other economic sources listed per record in data/raw/economic_footfall_raw.csv</li> <li>[4] All India Radio, Maha Kumbh concludes - <a href="https://www.newsonair.gov.in/maha-kumbh-mela-2025-concludes-in-prayagraj-on-maha-shivratri">https://www.newsonair.gov.in/maha-kumbh-mela-2025-concludes-in-prayagraj-on-maha-shivratri</a></li> <li>[5] NRF (16 Jan 2025); gov.cn (5 Feb 2025); GASTAT (5 Jun 2025); muenchen.de - global analogues</li> <li>[6] DataMeet India state boundaries (CC BY 2.5 IN) - <a href="https://github.com/datameet/maps">github.com/datameet/maps</a></li> </ol>                                                                                                                                                                                                        |